

DEPARTMENT OF MATHEMATICS AND STATISTICS

TITLE OF CLASS – ACSC/STAT 456-001 and Stat 862: Applied Stochastic Processes

Instructor: Dr. Andrei Volodin **Office:** CW 307.27

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Lectures: Monday, Wednesday, and Friday 12:30 – 1:20 p.m.

Location: ED 388 and by Zoom (see the link below)

Office Hours: Wednesday 9:00 - 11:00 a.m. or by an appointment

Web Page: <http://uregina.ca/~volodin/>

Zoom link for our class:

<https://uregina-ca.zoom.us/j/94810647841?pwd=OU50QWNFTm0rUkVBdFJJJa0VWbGZUUT09>

Meeting ID: 948 1064 7841

Passcode: 775937

CLASS DESCRIPTION – Topics include Markov Chains, Renewal Theory, and Brownian Motion.

Purpose of the Course – Stochastic processes play a basic role in several problems in many branches of actuarial, physical, and biological sciences, as well as in business, economics, and engineering. We will follow a heuristic and nonrigorous approach for studying stochastic processes to develop an intuitive feel for the subject which enables the student to ‘think probabilistically’.

PREREQUISITES – STAT 351.

TEXTBOOKS – Sheldon M. Ross. Introduction to Probability Models. Academic Press. The book is available through our university library for free. Tell me, if you cannot find it.

DETAILED DESCRIPTION – Markov Chains: Chapman-Kolmogorov equations, classification of states, limiting probabilities, the gambler’s ruin problem, mean time spent in transient state, branching process.

Poisson Process

Renewal Theory: Limit theorems, renewal reward processes, regenerative processes.

Brownian motion: Hitting times, maximum variable, geometric Brownian motion, drift, pricing stock options.

Martingales and Simulations (if time permits)

Grading Scheme: Your final grade will be calculated as follows:

Assignments: Not to hand in, just for you to prepare for the Midterm and Final Exam. I will post both the questions and solutions on URCourses.

Midterm (1): 40% in class on Monday, February 23 (right after the Winter break).

Final Exam: 60%